

Review

□ Maxwell's equations in free space

$$\varepsilon = \varepsilon_0; \mu = \mu_0; \text{ and } \rho = 0; J = 0$$

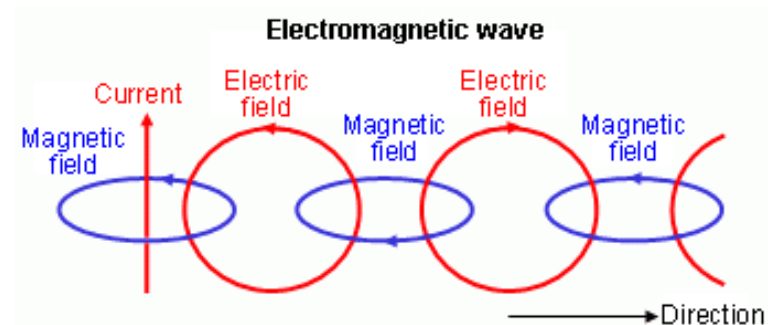
$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \varepsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

- *A time-varying magnetic field generates an electric field and a time-varying electric field generates a magnetic field.*



Maxwell's equations written in differential form

Based on Gauss's divergence theorem 高斯散度定理

Stokes's theorem 斯托克斯定理

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \epsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

$$\oiint_A \vec{F} \cdot d\vec{S} = \iiint_V \nabla \cdot \vec{F} dV$$

$$\oint_C \vec{F} \cdot d\vec{\ell} = \iint_A \nabla \times \vec{F} \cdot d\vec{S}$$



$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$



$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t}$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\frac{\partial B_y}{\partial t}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t}$$

$$\nabla \cdot \vec{E} = 0$$



$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

$$\nabla \cdot \vec{B} = 0$$



$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$



$$\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} = \mu_0 \epsilon_0 \frac{\partial E_x}{\partial t}$$

$$\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} = \mu_0 \epsilon_0 \frac{\partial E_y}{\partial t}$$

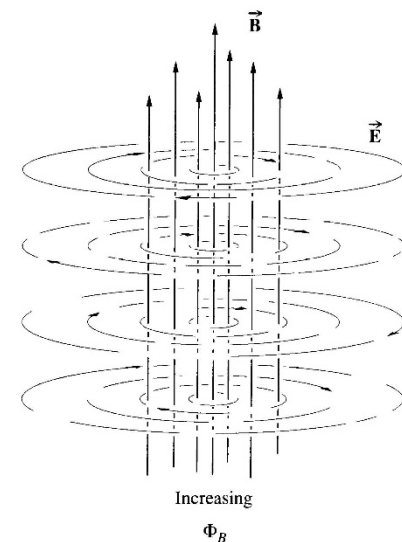
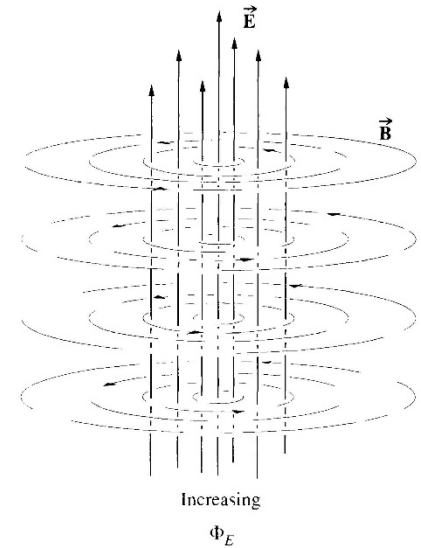
$$\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = \mu_0 \epsilon_0 \frac{\partial E_z}{\partial t}$$

2.2 Electromagnetic waves

Maxwell's equation tell us:

❑ A time-varying $\vec{E}$ -field generates a $\vec{B}$ -field, which is everywhere perpendicular to the direction in which $\vec{E}$ changes.

❑ A time-varying $\vec{B}$ -field generates a $\vec{E}$ -field, which is everywhere perpendicular to the direction in which $\vec{B}$ changes.



Maxwell's equations in the form of two vector expressions

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon} \quad \nabla \cdot \vec{B} = 0 \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{B}) &= \mu\sigma(\nabla \times \vec{E}) + \mu\epsilon\frac{\partial}{\partial t}(\nabla \times \vec{E}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{B}) &= \nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = -\nabla^2 \vec{B} \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla \times (\nabla \times \vec{B}) &= \mu\sigma(\nabla \times \vec{E}) + \mu\epsilon\frac{\partial}{\partial t}(\nabla \times \vec{E}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} \right\} \nabla^2 \vec{B} - \mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{E}) &= \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \right\} \nabla^2 \vec{E} - \mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} = \nabla\left(\frac{\rho}{\epsilon}\right)$$

In free space: $\epsilon = \epsilon_0$; $\mu = \mu_0$; and $\rho = 0$; $\sigma = 0$

$$\nabla^2 \vec{E} = \mu_0\epsilon_0\frac{\partial^2 \vec{E}}{\partial t^2} \quad \nabla^2 \vec{B} = \mu_0\epsilon_0\frac{\partial^2 \vec{B}}{\partial t^2}$$

Maxwell's equations for free space in the form of two vector expressions

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

The six components of the electromagnetic field $E_x, E_y, E_z, B_x, B_y, B_z$ obey the scalar differential wave equation

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$v = 1/\sqrt{\mu_0 \epsilon_0}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s} = c$$

This theoretical value was in remarkable agreement with the previously measured speed of light determined by Fizeau.

2.2.1 Transverse waves

- Consider the simple case of a plane wave propagating in vacuum in the positive x-direction.

$$\vec{E} = \vec{E}(x, t)$$

- According to Maxwell's equation:

$$\nabla \cdot \vec{E} = 0 \rightarrow \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

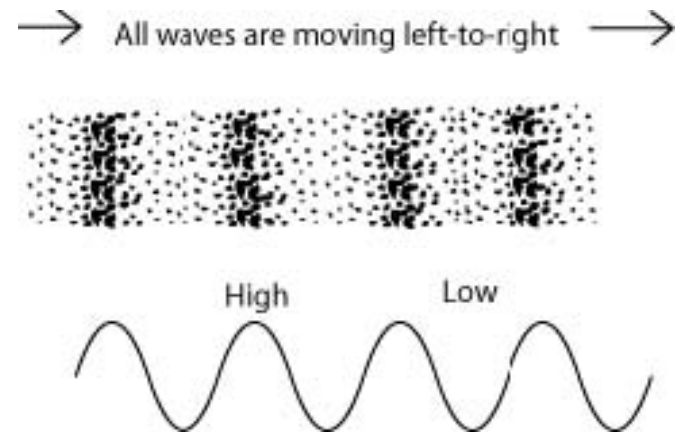
- We have

$$\frac{\partial E_x}{\partial x} = 0$$

- $E_x = \text{constant}$: it does not vary with x , so possibility cannot correspond to a traveling wave advancing in the positive x-direction.

- So $E_x = 0$

- The electromagnetic wave has no electric field component in the direction of propagation.



- ❑ To specify the wave, we first specify the moment-by-moment direction of $\vec{E}$.
- ❑ For simplify, we orient our coordinate axes to y-axis

$$\vec{E} = \hat{y}E_y(x, t)$$

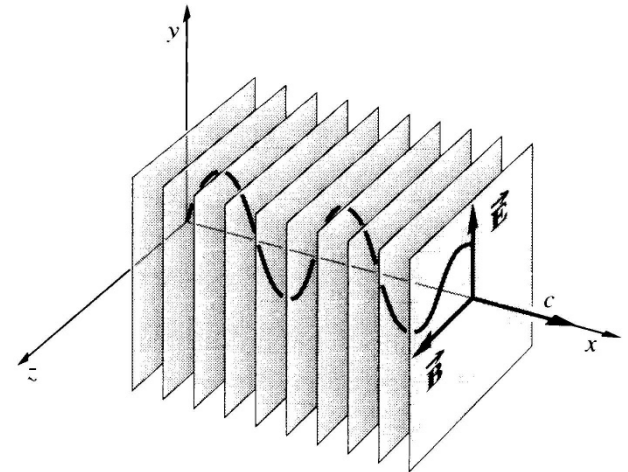
$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t}$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\frac{\partial B_y}{\partial t}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t}$$

$$\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t}$$

The time-dependent $\vec{B}$ -field can only have a component in the z-direction



- ❑ In free space, the plane electromagnetic wave is transverse.

Harmonic waves

$$E_y(x, t) = E_{0y} \cos \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right]$$

$$\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t}$$

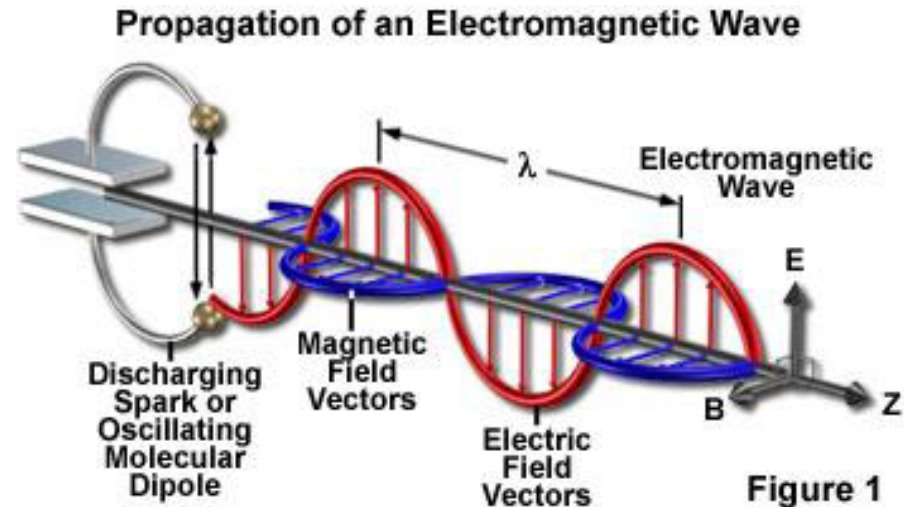
$$B_z = \int \frac{\partial E_y}{\partial x} dt$$

$$= -\frac{E_{0y}}{c} \int \sin \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right] dt$$

$$= \frac{1}{c} E_{0y} \cos \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right]$$

$$= \frac{E_y}{c}$$

- $\vec{E}$ and $\vec{B}$ have the same time-dependence;
- $\vec{E}$ and $\vec{B}$ are in-phase at all points in space;
- $\vec{E}$ and $\vec{B}$ are mutually perpendicular;
- $\vec{E} \times \vec{B}$ points in the propagation direction.



2.3 Energy: The Poynting Vector

- Any electromagnetic wave exists within some region of space, and the field has energy it is a thing-in-itself.

Energy density (u): the radiant energy per unit volume

The energy density of the $\vec{E}$ -field:

$$u_E = \frac{\epsilon_0}{2} E^2$$

The energy density of the $\vec{B}$ -field:

$$u_B = \frac{1}{2\mu_0} B^2$$

Using $c = 1/\sqrt{\mu_0\epsilon_0}$, we have $u_E = u_B$

The total energy density of electromagnetic fields

$$u_{total} = u_E + u_B = \epsilon_0 E^2 = \frac{1}{\mu_0} B^2$$

The transport of energy per unit time across a unit area.

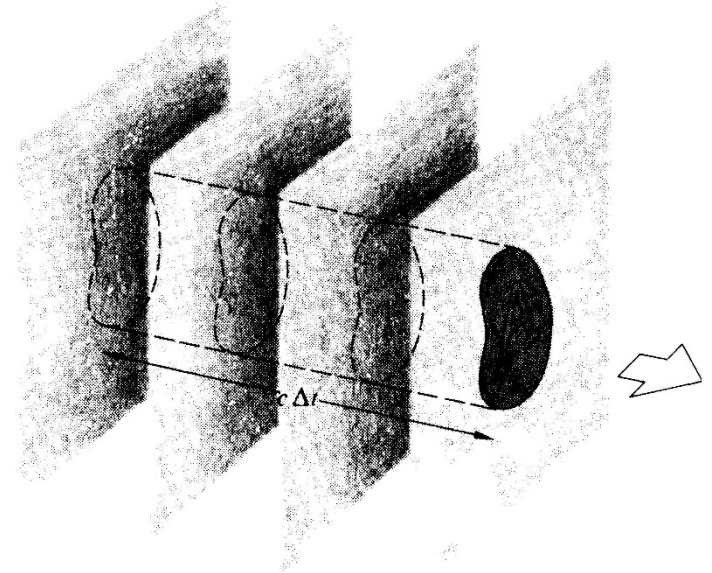
$$S = \frac{uc\Delta tA}{\Delta tA} = uc = \frac{1}{\mu_0} EB$$

Vector $\vec{S}$: energy flowing in the direction of the propagation of the wave

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$\vec{S} = c^2 \epsilon_0 \vec{E} \times \vec{B}$$

The Poynting vector: the magnitude of $\vec{S}$ is the power per unit area crossing a surface whose normal is parallel to $\vec{S}$

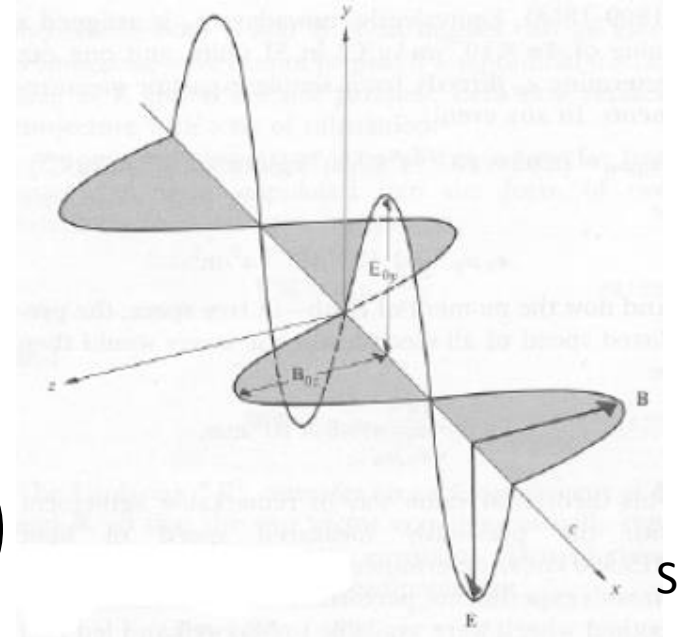


- Consider a harmonic, linearly polarized plane wave traveling through free space in the direction of $\vec{k}$

$$\vec{E} = \vec{E}_0 \cos(\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{B} = \vec{B}_0 \cos(\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{S} = c^2 \epsilon_0 \vec{E}_0 \times \vec{B}_0 \cos^2(\vec{k} \cdot \vec{r} - \omega t)$$



This is the instantaneous flow of energy per unit area per unit time.

Irradiance and power

The average energy per unit area per unit time (irradiance I):

$$\begin{aligned} I = \langle S \rangle_T &= c^2 \epsilon_0 \left| \vec{E}_0 \times \vec{B}_0 \right| \left\langle \cos^2 \left(\vec{k} \cdot \vec{r} - \omega t \right) \right\rangle \\ &= \frac{c^2 \epsilon_0}{2} \left| \vec{E}_0 \times \vec{B}_0 \right| = \frac{c \epsilon_0}{2} E_0^2 \end{aligned}$$

The irradiance is proportional to the square of the amplitude of the electric field.

E is considerably more effective at exerting forces and doing work on charges than is B, we refer to E as the optical field and use the above equation almost exclusively.

The optical power (P): the time rate of flow of radiant energy (unit watts):

$$I = \frac{P}{S}$$

The inverse square law

From the solution of the differential wave equation
Spherical wave function:

$$E(r, t) = \left(\frac{\mathcal{A}}{r} \right) \cos(kr \mp \omega t)$$

From the energy conservation

$$I_1 \pi r_1^2 = I_2 \pi r_2^2$$

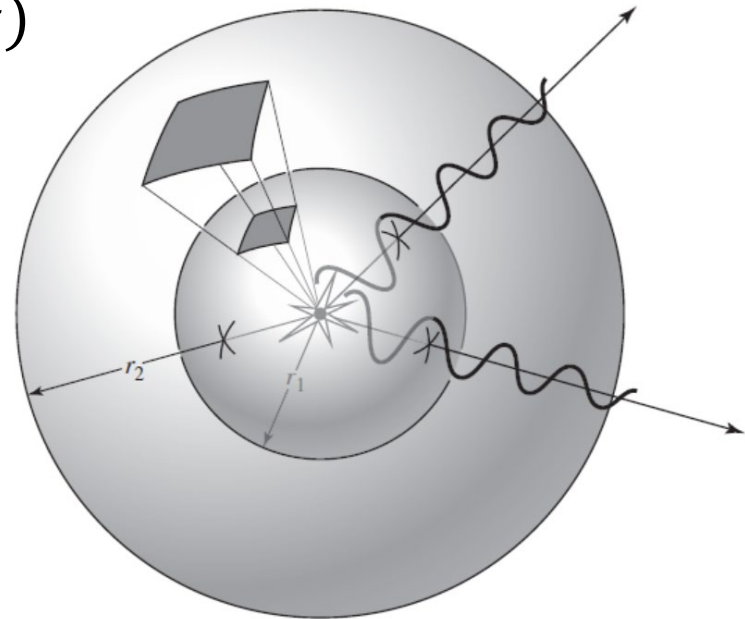
We get

$$r_1 E_0(r_1) = r_2 E_0(r_2)$$

As to arbitrary r

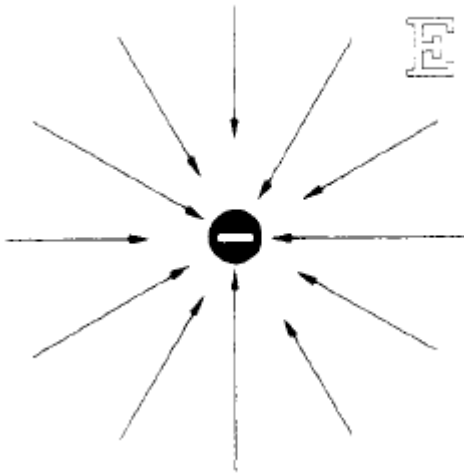
$$r E_0(r) = \text{constant}$$

$$I \propto \frac{1}{r^2} \quad \text{Inverse square law}$$



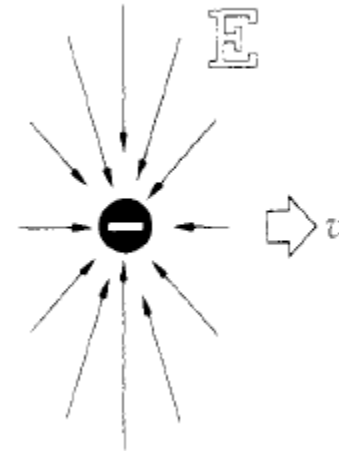
2.4 Radiation

A stationary charge: constant $\vec{E}$ -field, no $\vec{B}$ -field, **no radiation**.



Electric field of a stationary electron

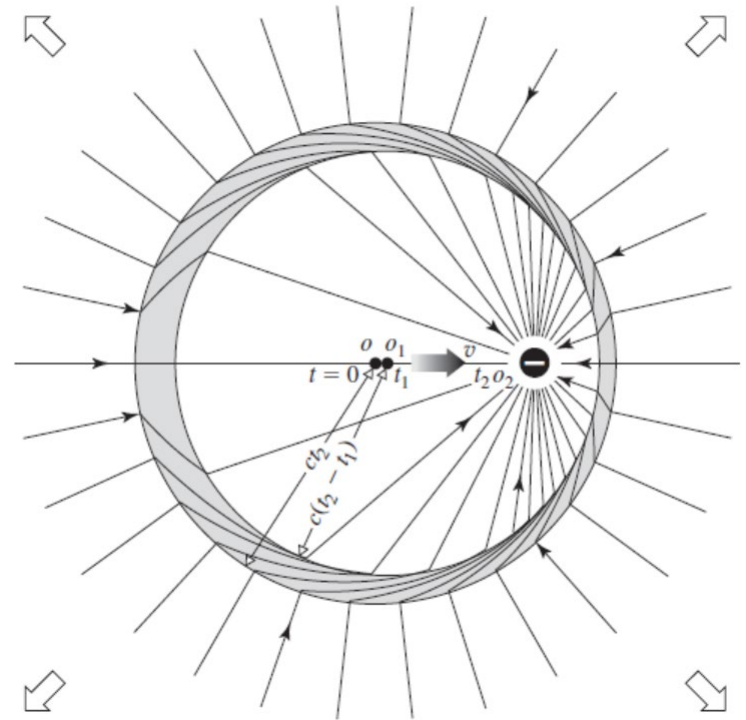
A uniformly moving charge: constant $\vec{E}$ -field and $\vec{B}$ -field, but **no radiation**.



Electric field of a moving electron
(v is comparable with c)

A nonuniformly moving charge: **radiation**.

- ❑ From this kinked region we see that there exists a transverse component of the electric field, $\mathbf{E}_T$.
- ❑ This transverse component propagates outward, away from the charge, at the speed of light.
- ❑ Thus, the pulse which contains the kink is not only a function of space, but also of time.
- ❑ From Maxwell's equations, we see that the transverse field is therefore accompanied by a magnetic field.



2.5 Photon

❑ The electromagnetic radiation is quantized and exists in the elementary amounts that we call **photons**.

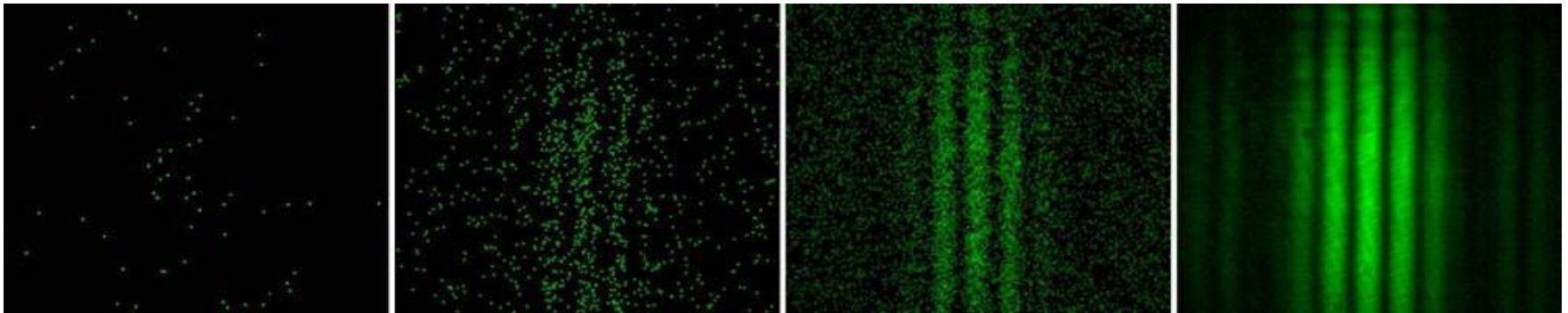
❑ The energy of single photon can be expressed as $E=h\nu$

Irradiance is proportional to the number of photon.

Quantum picture of light



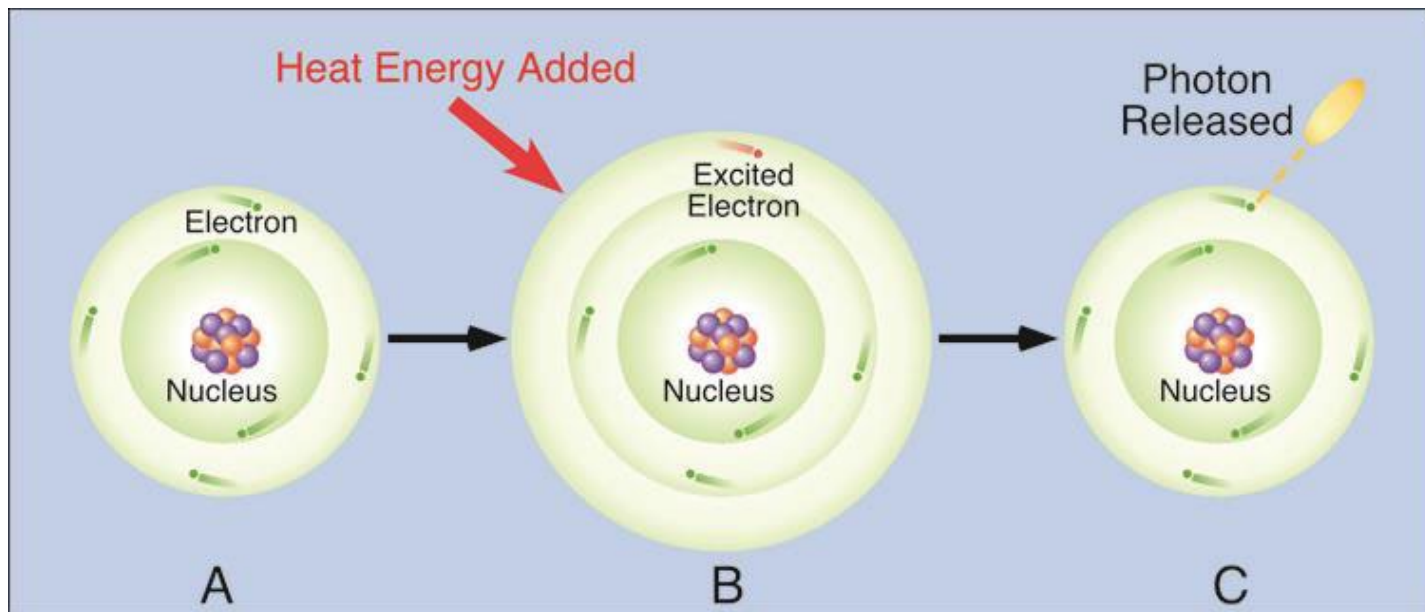
its wave property



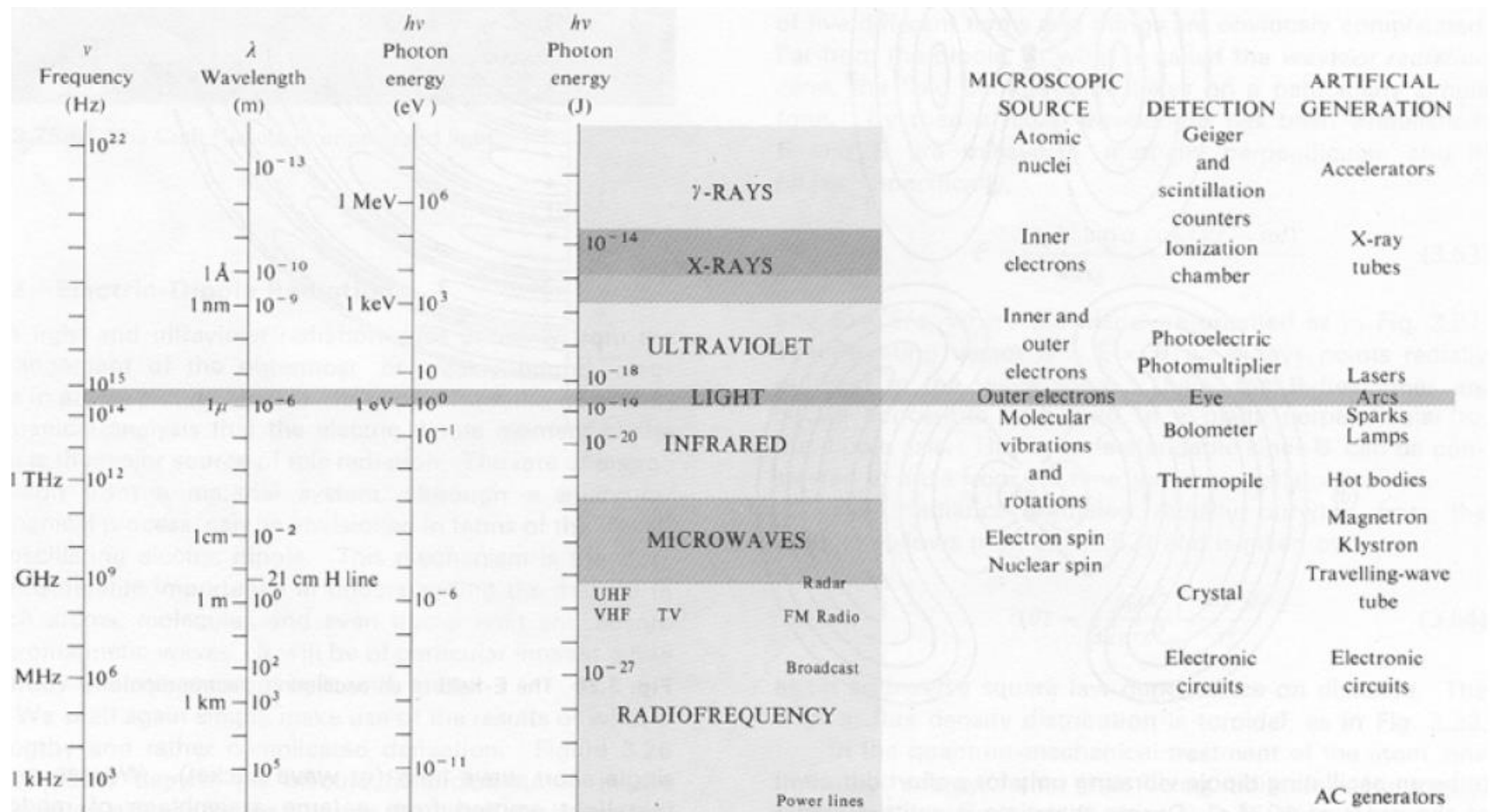
Single-photon camera recording of photons from a double slit illuminated by very weak laser light. Left to right: single frame, superposition of 200, 1'000, and 500'000 frames.

Emission of light from atoms

- ❑ (A) Without any disturbance, every electron is in the lowest possible energy state (**ground state**) available to it.
- ❑ (B) After absorbing a blast of energy, one or more electrons are excited to a higher energy level (**excited state**).
- ❑ (C) The excited state is unstable and temporary, electron drops back and returns to the ground state, with the emission of a **photon**.



2.6 The electromagnetic spectrum



2.7 Light in bulk matter

□ Objective: dielectric or nonconducting materials

The phase speed in the medium

$$v = 1/\sqrt{\epsilon\mu}$$

Absolute index of refraction (n):

$$n \equiv \frac{c}{v} = \frac{\sqrt{\epsilon\mu}}{\sqrt{\epsilon_0\mu_0}}$$

Consider the nonmagnetic materials

$$n = \sqrt{\frac{\epsilon}{\epsilon_0}}$$

The dependence of n on the wavelength of light is a well-known effect called **dispersion**.



Newton's prism

Scattering and absorption

- ❑ **Case I: the photon's energy matches that of one of the excited states.**
- ❑ As to the atom, the light will be absorbed, making a quantum jump to that higher energy level.
- ❑ In the dense atomic landscape of ordinary gases, solids and liquids, it is very likely that this excitation energy will rapidly be transferred, via collisions, to random atomic motion, thermal energy, before a photon can be emitted. It is referred to as **dissipative absorption (损耗吸收)**.

- ❑ **Case II: the photon's frequency or energy is lower than the resonance frequency.**
- ❑ The electromagnetic field of the light can drive the electron cloud into oscillation.
- ❑ Once the electron cloud starts to vibrate with respect to the positive nucleus, the system constitutes **an oscillating dipole** and will immediately begin to radiate at that same frequency.
- ❑ The resulting scattered light consists of a photon that sails off in some direction carrying the same amount of energy as did the incident photon—**the scattering is elastic.**